# AI Speech and language variation & change

A study on manifestation of AI speech in human natural language

class: [16838] germanische sprachen im vergleich
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## Preliminary

inspired by BSI (2025) and Yakura et al. (2025) which found evidence for GPT influenced human language after the introduction of chatGPT we tried to replicate the Yakura et al. (2025) pipeline of building an AI vocabulary (model preferred lemmata) and compare frequencies of model typical words across pre- and post model introduction human language corpora. The first draft essai proves their hypothesis that LLM generated language manifests within human natural language.

## Questions / Hypotheses

- do humans adapt to language produced by LLM and incorporate model vocabulary into their own language production?
- H1: we will find higher frequencies of LLM typical vocabulary within human natural language corpora after onset of model introduction.

## Methods / Data

our human languaga data consists of raw texts from german bundestag plenary protocols, DIP (2026). the LLM corpus consists of model summaries of a first subset of these texts generated with the prompt (to gemini API) you see in C1.

we first devised model typical lemmata in the AI corpus which are distinct for that corpus using the log odds score calculated over pre/post target.

```
C1: System prompt: You are a member of german
parliament. Prepare a summary of the text
provided to present at a local community meeting
of your party members. Output in german language,
no preamble, no extra information, just the plain
text. Wordcount maximal 300 words, containing not
more than 5% of the keywords of the text provided
and explicitly not just a list of keywords but an
entertaining text. You are supposed to interprete
freely, including background insights on daily
politics. Keep in mind thatthe text will be used
as is as keynotes to the talk being held to the
locals. Text:
```

## p2

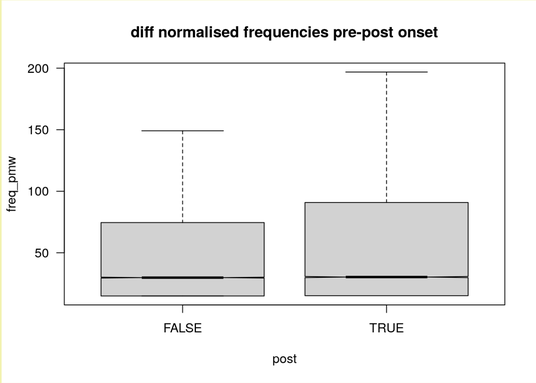


Figure 1: stats descriptive

## p3

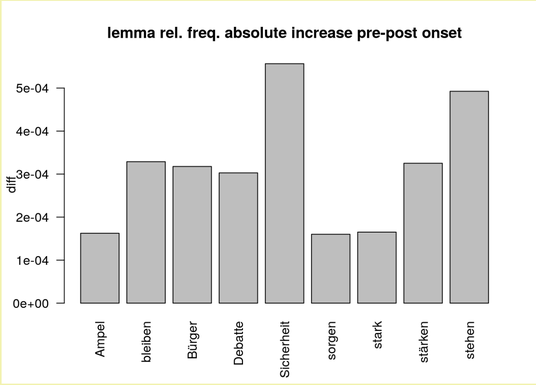


Figure 2: lemma frequency increase

```
corpus stats
    stat                  var      pre     post
1   mean freq_rel gpt lemmas 0.000704 0.000751
2   mean freq normalised pmw   29.825   30.288
3 tokens                   n  1640446  1940385
```

## c3: regression simple model

```
Call:
lm(formula = freq_pmw ~ post + gp, data = df_lg.norm)

Residuals:
   Min     1Q Median     3Q    Max
 -1400   -435   -388   -243 436922

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   833.75      38.64  21.576  < 2e-16 ***
postTRUE       83.94      20.64   4.067 4.77e-05 ***
gp             68.11       5.20  13.097  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3765 on 133087 degrees of freedom
  (1393 observations deleted due to missingness)
Multiple R-squared:  0.0014,    Adjusted R-squared:  0.001385
F-statistic: 93.32 on 2 and 133087 DF,  p-value: < 2.2e-16

Analysis of Variance Table

Response: freq_pmw
              Df     Sum Sq    Mean Sq F value    Pr(>F)
post           1 2.1394e+08  213944737  15.096 0.0001022 ***
gp             1 2.4310e+09 2431030842 171.537 < 2.2e-16 ***
Residuals 133087 1.8861e+12   14172027
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

## p4

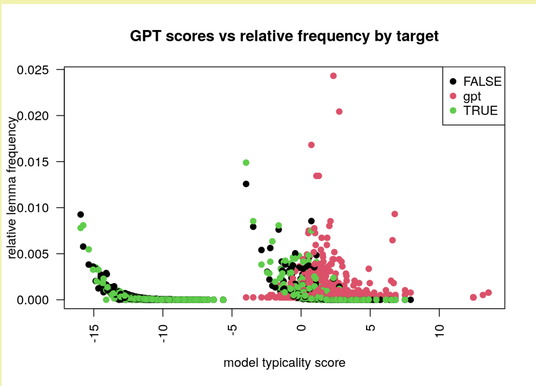


Figure 3: model vocab distribution

## Results

to gather an insight, yet with simple descriptive stats comparing the raw frequencies of gpt-preferred lemmas in pre- and post-gemini onset we find that in the target corpus the occurences of these lemma increase, see Figure 1.

the 9 lemma where 1. the relative frequency of lemma in target=post is higher than in target=pre (n=27) and 2. that frequency exceeds the mean relative frequency is displayed in Figure 2

{placeholder}{we also found significant frequency increases using a linear regression model, but after experiencing issues with udpipe which first become visible when revisiting our data we had to lemmatize corpus again using the stanza nlp pipeline. results will show up here when evaluated with new lemmas.}

## c4: regression mixed effects model

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: freq_pmw ~ post + gp + (1 | lemma)
   Data: df_lg.norm

REML criterion at convergence: 2490672

Scaled residuals:
    Min      1Q  Median      3Q     Max
-48.624  -0.047  -0.032   0.008  74.360

Random effects:
 Groups   Name        Variance Std.Dev.
 lemma    (Intercept) 8839356  2973
 Residual             1754704  1325
Number of obs: 133090, groups:  lemma, 99649

Fixed effects:
              Estimate Std. Error        df t value Pr(>|t|)
(Intercept) 4.344e+02  3.966e+01 1.055e+05  10.953   <2e-16 ***
postTRUE    1.246e+02  9.496e+00 5.602e+04  13.126   <2e-16 ***
gp          2.932e+01  5.723e+00 1.017e+05   5.124    3e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
         (Intr) psTRUE
postTRUE -0.110
gp        0.959  0.009

Type III Analysis of Variance Table with Satterthwaite's method
        Sum Sq   Mean Sq NumDF  DenDF F value    Pr(>F)
post 302319616 302319616     1  56015 172.291 < 2.2e-16 ***
gp    46071208  46071208     1 101716  26.256 2.996e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

## references

BSI, Brand Science Institute. 2025. “Wie KI Unsere Sprache Verändert – Eine Empirische Studie.” https://www.bsi.ag/cases/104-case-studie-wie-ki-unsere-sprache-veraendert---eine-empirische-studie.html.

DIP. 2026. “DIP - Bundestagsprotokolle.” Docs. *DIP - API*. Berlin. https://dip.bundestag.de/%C3%BCber-dip/hilfe/api#content.

Yakura, Hiromu, Ezequiel Lopez-Lopez, Levin Brinkmann, Ignacio Serna, Prateek Gupta, Ivan Soraperra, and Iyad Rahwan. 2025. “Empirical Evidence of Large Language Model’s Influence on Human Spoken Communication.” arXiv. https://doi.org/10.48550/arXiv.2409.01754.